LITTLE STREAM DETECTOR

LSD 3.0 Technical Documentation

Capabilities, operation and result interpretation

Property	Value
Release	3.0
Edition	User-focused final edition with native VVC and AV2 support
Platform	Windows PowerShell 5.1 + .NET Framework
External multimedia tools	None required
Primary use	Native stream inspection and A/B comparison
Revision	11 October 2026

Purpose of this revision Documents native AV2 parsing, two-pass frame-event resolution, normative Base Q Index analysis, centralized histogram reporting, and the supported AV2 container scope.

1. What LSD 3.0 Does

Little Stream Detector 3.0 is a native Windows media inspection application for examining video files without launching FFmpeg, FFprobe, MediaInfo, or another external multimedia executable. It reads supported containers, identifies tracks, analyzes supported elementary streams, and presents human-readable and structured results.

Typical uses

Inspect a single file, compare an encode with a reference, verify stream structure after remuxing, examine bitrate behavior, compare frame-level quantizer distributions, and validate native VVC or AV2 picture structure.

Core capabilities

- Open a video from the file picker or by drag and drop.
- Analyze container, track, codec, timing, bitrate, frame-level information, and pixel format where supported.
- Keep the latest analysis as A Current and optionally compare it with an immutable B Reference.
- Display independent Summary A and Summary B reports, Streams data, JSON, and a processing Log.
- Plot A/B bitrate profiles and display codec-specific QP or quantizer distributions.
- Report unavailable information as N/A rather than estimating a value.
- Analyze supported AV2 streams through the same canonical media pipeline used by the other codec analyzers.

Supported families

Area	Supported families
Containers	Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV
Video	H.264/AVC, H.265/HEVC, H.266/VVC, AV1, AV2, VP8, VP9, MPEG-4 Part 2/XviD
Audio	AAC, MP3, PCM, AC-3, E-AC-3
Outputs	Summary A/B, Streams, JSON, Log, bitrate profile, quantizer distribution

2. Operating the Application

Basic workflow

1. Choose Open video or drop a supported media file into the drop area.
2. Wait for analysis to complete. The current result becomes A Current.
3. Review Summary A, Streams, JSON, and Log as needed.
4. Inspect the bitrate profile and quantizer panel.
5. To compare files, retain a result as B Reference and analyze the other file as A Current.
6. Use Remove Ref B when the reference is no longer needed.
7. Use Copy to copy the active report.

A/B comparison model

- A Current is the most recently analyzed result and is shown in blue.
- B Reference is a stored comparison result and is shown in gold.
- The interface supports one current result and one reference result at a time.
- Percentages are useful when files have different frame counts; raw counts show exact frequency.

3. Understanding the Reports

View	What it provides
Summary A	Readable report for the current file.
Summary B	Readable report for the stored reference.
Streams	Structured information about native tracks.
JSON	Serializable metadata and stream information.
Log	Processing lifecycle, validation notes, and first-failure messages.

Information commonly shown

- File name, modification time, size, and duration.
- Detected container and native document type.
- Number and type of native tracks.
- Video dimensions, codec family, pixel format, and codec-specific configuration.
- Audio format, rates, channels, and bitrate where available.
- Canonical analysis status and routing information.
- Frame types, keyframe information, and codec-dependent quantizer statistics where analysis is complete.

AV2 report structure

AV2 uses its own parser and data model, but the Summary follows the same presentation style as the mature AV1 report. The AV2 report is divided into Sample / OBU Analysis, Frame Analysis, Reference / Geometry, Base Q Index Analysis, Native Engine, and Quantizer Analysis sections.

Separate parser, consistent presentation AV1 and AV2 do not share codec syntax parsing. They only share the centralized statistics, table, graph, and A/B presentation layers after each parser publishes its own validated histogram.

4. Bitrate Profile

The bitrate profile plots the amount of video data over presentation-time intervals. A Current is blue and B Reference is gold. This makes local peaks, quiet sections, and differences between two files visible over the full duration.

Reading the graph

- Tall bars indicate intervals containing more encoded data.
- Low bars indicate intervals containing less encoded data.
- Peak bucket reports the largest interval value observed for the active file.
- The horizontal position corresponds to playback time.
- Hovering displays the selected interval and available A/B bitrate values.

Do not judge quality from bitrate alone A higher instantaneous or average bitrate does not automatically mean higher visual quality. Codec family, encoder settings, source complexity, and quantizer behavior must be considered together.

5. QP and Quantizer Distribution

The quantizer panel summarizes the codec-specific quantizer metric for analyzed frames. AVC, HEVC, and VVC use frame-level SliceQPY when complete native parsing succeeds. Other codecs use the corresponding native quantizer index.

Codec family	Histogram value
H.264 / AVC	Frame-level SliceQPY
H.265 / HEVC	Frame-level SliceQPY
H.266 / VVC	Frame-level SliceQPY
AV1	Base Q Index
AV2	Base Q Index
VP8	Base Q Index
VP9	Base Q Index
MPEG-4 Part 2	VOP quantizer

Centralized presentation

Each codec parser produces its own validated values. Those values are serialized into a codec-specific histogram and passed to the shared statistics and rendering layer. The common layer calculates Frames, Avg, Median, Mode, and Range and draws the distribution and A/B comparison.

Statistics above the distribution

Statistic	Meaning
Frames	Number of events contributing to quantizer analysis.
Avg	Arithmetic mean of analyzed values.
Median	Middle quantizer value after sorting.
Mode	Most frequently occurring quantizer value.
Range	Lowest and highest recorded values.

Percentage and count modes

Click the fixed QP distribution header to switch A/B columns between percentage mode and count mode. The colored bars remain a relative visual representation and A/B columns stay aligned.

6. Interpreting Quantizer Results

- A narrow distribution indicates many frames were encoded within a relatively small quantizer range.
- A broad distribution indicates stronger variation in quantization across the file.
- A different mode shows that the most common frame quantizer changed.
- For an unchanged elementary stream after a lossless remux, frame counts and quantizer histograms should normally remain equivalent.

Important comparison rule A and B distributions are directly comparable only when both files use the same quantizer metric. Different codec families are not a one-to-one quality scale.

What QP does not prove by itself

- QP alone does not measure visible quality.
- QP alone does not establish encoder efficiency.
- The same numeric value can have different practical meaning across codecs or encoding structures.
- LSD does not decode pixels and does not calculate PSNR, SSIM, or VMAF.

7. Codec and Audio Analysis

Video information

Family	Native analysis
H.264 / AVC	SPS/PPS/VUI, I/P/B, pixel format, and SliceQPY
H.265 / HEVC	VPS/SPS/PPS, I/P/B, pixel format, and SliceQPY
H.266 / VVC	vvcC, SPS/PPS, two-pass NAL and embedded-PH traversal, I/P/B, pixel format, and SliceQPY
AV1	OBU traversal, frame state, geometry, pixel format, and Base Q Index
AV2	OBU traversal, immutable frame-event index, two-pass state resolution, geometry, and Base Q Index
VP8	Frame tag, KEY/INTER, and Base Q Index
VP9	Superframes, frame state, references, and Base Q Index
MPEG-4 Part 2 / XviD	VOL/VOP, I/P/B/S, GMC, and VOP quantizer

Native AV2 analysis

- The container adapter publishes bounded canonical video samples to the common media pipeline.
- Pass 1 traverses AV2 OBU framing, records OBU inventory, parses Sequence Header data, and builds an immutable frame-event index.
- Pass 2 resolves CLK, OLK, and regular frame events in sequence; regular events use explicit or inherited state as required.
- The Base Q Index timeline is published only when event accounting and structural validation succeed.
- The AV2 histogram is produced by the AV2 parser and passed to the same centralized statistics and graph layer used by other codecs.
- Film Grain OBUs are routed independently from frame-header syntax. Unsupported active syntax paths fail closed rather than being estimated.

AV2 Summary sections

Section	Purpose
Sample / OBU Analysis	Container samples, OBU counts, frame carriers, event relation, and OBU validation.
Frame Analysis	Event accounting, CLK/OLK/regular routing, explicit/inherited resolution, and rejected events.
Reference / Geometry	Reference slots, maximum dimensions, superblock and tile limits.
Base Q Index Analysis	Resolved event count, average, median, mode, range, completeness, and validation.

Audio information

Family	Validation and reporting
AAC	Configuration, core/output rate, channels, SBR, PS, payload, and bitrate
MP3	Version/layer, bitrate, sample rate, padding, and frame size

AC-3 / E-AC-3	Syncword, bitstream ID, rate, frame size, blocks, and channels
PCM	Block alignment, sample format, duration, and bitrate

8. Support Matrix and Limitations

Video	Matroska/WebM	MP4/MOV	fMP4	AVI
H.264 / AVC	Yes	Yes	Yes*	Yes, Annex B
H.265 / HEVC	Yes	Yes	Yes*	No
H.266 / VVC	Yes**	Yes	Yes*	No
AV1	Yes	Yes	Yes*	No
AV2	Yes***	Not currently published	Not currently published	No
VP8	Yes	No	No	No
VP9	Yes	Yes, vp09	Yes, vp09*	No
MPEG-4 Part 2	Yes	Yes	Yes*	Yes

* Fragmented MP4 reuses an existing codec analyzer when the sample entry and payload format are supported.

** VVC requires a supported length-prefixed NAL representation and valid codec-private context.

*** AV2 support currently targets Matroska/WebM V_AV2 streams and uses native in-memory OBU processing.

Known limitations

- Encrypted CENC samples and auxiliary encryption structures are not supported.
- Standalone media segments without an initialization moov box are not supported.
- Files still being written and live network streams are not supported.
- VP8 is supported only in Matroska/WebM; VP9 in AVI is not supported.
- AVI H.264/AVC support is intended for Annex B streams.
- Container-only color metadata is not used as a fallback.
- VVC results are published only after complete two-pass validation.
- AV2 support is focused on structural stream analysis and Base Q Index extraction; LSD does not decode AV2 pixels.
- AV2 syntax paths that are not fully supported are rejected rather than estimated. Runtime validation of optional branches depends on suitable source samples.
- Quantizer distributions are directly comparable only when A and B use the same quantizer metric.

9. Validation, Troubleshooting and Good Practice

When an analysis is incomplete

1. Open Log and identify the first validation or parser failure.

2. Confirm that the file is complete and readable from local storage.

3. Check whether the container/codec combination is supported.

4. Compare analyzed event count with expected duration and frame rate.

5. Treat disabled or N/A histograms as unavailable rather than zero-valued results.

6. For AV2, check OBU validation, frame-event accounting, rejected second-pass events, guarded resolutions, and Base Q completeness.

Useful expectations for a lossless remux

Check	Expected relationship

Video/audio sample count	Equal, or differences explained by container edits.
Payload bytes	Equivalent elementary-stream accounting.
Keyframe and frame-event counts	Equal for unchanged elementary streams.
Quantizer histogram	Equal for unchanged elementary streams using the same metric.
Pixel format and bit depth	Equal when codec sequence information is unchanged.
Rejected samples/headers	Zero for a validated asset.

Appendix A. Concise Technical Reference

LSD uses PowerShell for orchestration and the WinForms interface, with embedded C# for native parsing and analysis. Supported containers are converted into a shared, bounded sample model before codec and audio analysis. Standard MP4 and fragmented MP4 use separate indexing routes but feed the same reporting, bitrate, and A/B comparison layers.

Processing flow

Pipeline File selection -> container detection -> sample indexing -> canonical media model -> codec/audio analysis -> validation -> reports and A/B graphs

AV2 processing flow

AV2 route Canonical video samples -> AV2 OBU collection -> immutable frame-event index -> sequential Resolve pass -> validated Base Q timeline -> AV2 histogram -> shared statistics and graph layer

Defensive behavior

- Sample ranges must remain inside the source file.
- Unavailable information is reported as N/A.
- Quantizer histograms remain unavailable when parser completeness checks fail.
- AV2 Base Q Index values are published only from a complete validated event timeline.
- Optional AV2 syntax paths fail closed when required state or syntax is unavailable.
- No external multimedia tools or temporary analysis files are required.

Terminology

Term	Definition
A Current	Most recently analyzed result.
B Reference	Stored immutable comparison result.
Canonical sample	Container-neutral bounded sample with timing and key state.
QP / quantizer distribution	Frequency distribution of the codec-specific quantizer metric.
AV2 frame event	Indexed CLK, OLK, or regular coded event resolved from AV2 OBU carriers.
AV2 two-pass analysis	OBU/event collection followed by context-aware sequential state and Base Q resolution.
pcs mode	Raw per-quantizer frame counts for A and B.
fMP4	Fragmented MP4 whose media samples are described by movie fragments.

10. Manual Preview

Feature update - 23 October 2026

The Preview tab provides two independent, muted panes for manual visual comparison of A Current and B Reference. Preview is a display aid only and remains separate from LSD's native container and bitstream analysis.

Operation

• **Ref A** uses a blue active-state accent and follows A Current.

• **Ref B** uses a gold active-state accent and follows B Reference.

• One shared preview implementation creates two separate Windows media instances. Each pane has its own source, state, controls, and abstract axis.

• The axes are independent and are not driven by the source video's reported frame rate.

• Click a pane to select it. Left Arrow and Right Arrow move the active pane by one axis step; Space toggles Play/Pause.

• The picture area, including black letterbox regions, participates in pane selection.

Availability and interpretation

• If a source is missing or cannot be decoded by the Windows media pipeline, the pane displays **PREVIEW UNAVAILABLE**.

• A file may still be fully analyzable by LSD even when Preview is unavailable, because native analysis does not depend on pixel decoding.

• Preview does not synchronize panes, determine offsets, or infer frame-rate conversion, editing, interpolation, or source-frame correspondence.

• Equal axis positions do not prove that the displayed pictures originate from the same source frame. The user manually selects the pictures to compare.

Layer	Role
Native LSD analysis	Deterministic container and bitstream inspection, reports, bitrate profiles, and quantizer statistics.
Manual Preview	Two independent Windows-decoded display panes controlled by separate abstract axes.